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MAT-AGCA: Multi Augmentation Technique on Small Dataset for Balinese Character Recognition Using Convolutional Neural Network

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Abstract

The *lontar* manuscript is an ancient Balinese cultural heritage written using Balinese character on palm leaves. The recognition of Balinese characters in *lontar* is challenging because it has noise and limited data availability. To solve these problems, data augmentation is needed to increase the variety and amount of data to improve recognition performance. In this study, we collected Balinese character images from 50 *lontar* manuscript writers. We proposed MAT-AGCA that combines Adaptive Gaussian Thresholding and Convolutional Autoencoder for data augmentation. Based on experiments using InceptionResnetV2, DenseNet169, ResNet152V2, VGG19, and MobileNetV2, our proposed method achieved the best performance with 96.29% accuracy.
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Keywords: Balinese Character, *Lontar* manuscript, Data Augmentation, Adaptive Gaussian Thresholding, Convolutional Autoencoder.

1. Introduction

The Balinese characters is derived from the Brahmi script of ancient India. Balinese characters consists of the *Wianjana*, *Suara*, *Pengangge*, and *Gempelan*. Balinese characters can be found in all regions of Bali island, starting from the street name, office signpost, and *lontar* manuscripts. *Lontar* manuscript is an ancient Balinese cultural heritage written on palm leaves using the Balinese characters. *Lontar* has a close relationship in religious life in Bali. Several attempts have been made by previous researchers to develop Balinese characters recognition system, but there are still problems that become challenges in recognizing Balinese characters in *lontar* manuscripts. In this study, we focus on proposing an augmentation method suitable for a set of handwritten images with limited number, high variation strokes and noise, to improve the recognition performance of Balinese characters. *Lontar* manuscripts are written on dried palm leaves using *Pengrupak*, which is a small knife specically used for writing the Balinese characters. The use of *Pengrupak* causes the writing of the Balinese characters to be inconsistent and have various writing variations. The use of palm leaves is also very influential because the palm leaves have fiber or traces of candlenut soot to distinguish the characters from palm leaves. In character recognition, noise becomes an important factor that influences recognition results. Noise reduction technique plays an important role in separating Balinese characters from the background. In Balinese characters recognition on *lontar* manuscript, *lontar* leaf fibers become noise, which affects the recognition process. Besides, there are still limited variations in the Balinese character dataset to handle variations in Balinese characters writing in lontar manuscripts, therefore an augmentation technique is needed to provide writing variations.

Research on object recognition has been carried out in various domains. Previous studies applied deep learning to cultural object recognition [1] [2]. Satheesh and Raj [3] proposed Multiple Kernel Fuzzy C Means (MKFCM) and hybrid classifiers to differentiate the images as Cavitary TB and Miliary TB using nomination method of classification. Related research on character recognition has been done on the Balinese and Javanese characters. Darma [4] proposed Zoning and k-Nearest Neighbors on Balinese character recognition. Related research has

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been conducted by Sugianela and Suciati [5] by applying a histogram of oriented gradient on the recognition of Javanese character. Ramteke et al. [6] proposed a method for optical character recognition and classification using Support Vector Machine and Adaptive Cuckoo Search (SVM-ACS) algorithm for Marathi text document. Related research has been conducted by Prasetyo et al. [7] by proposing a cosine K-Nearest Neighbor in Milkfish Eye Classification. This method uses a recent weighting scheme where the weights were no longer gained from the nearest neighbor individually, but by involving all pair of the nearest neighbor. These studies have not applied the preprocessing technique as data augmentation to improve recognition performance.

In this study, we use *Palalubangan lontar* manuscripts and collected Balinese characters from 50 *lontar* manuscript writers in Bali. We contribute to this research topic by presenting a method for data augmentation of Balinese characters on *lontar* manuscript. In this work, our research aims to overcome the limited number of Balinese characters dataset in recognizing Balinese characters in the *lontar* manuscript by proposing MAT-AGCA as a data augmentation technique. MAT-AGCA can produce new Balinese character images synthetically with better image quality as the new dataset, namely Handwritten Balinese Characters of *lontar* (HBCL) dataset as a solution for a limited number of Balinese characters. We developed a classification model to recognize Balinese characters in *lontar* manuscript images using Convolutional Neural Network architecture, which is the current best method for image recognition.

This paper is organized as follows: Section 2 discusses the literature reviews and related works. Section 3 described the Balinese characters on *lontar* manuscript and presents our proposed method for data augmentation using MAT-AGCA on the Balinese characters recognition. Section 4 shows the result of experimental study and discussions. Finally, Section 5 presents the conclusions and future work.

2. Literature review

Data augmentation can provide data variation and scale up the dataset to improve recognition performance. Preprocessing techniques can be used to add data variations by improving image quality. In Balinese characters recognition, the *lontar* manuscript has a noise that needs to be reduced before character recognition. Various studies applied various preprocessing methods to improve image quality. Dnyandeo and Nipanikar [8] applied the adaptive thresholding technique for vehicle license plate recognition. This study uses four adaptive thresholding techniques, i.e., mean, median, gaussian filter, and integral image. The research discusses the importance of adaptive thresholding in light illumination and resolution problems. Darma and Sutramiani [9] applied preprocessing of Balinese characters based on color space using CIELab. Related research on preprocessing conducted by Sudarma and Sutramiani [10] on Balinese script on papyrus. Related research on image quality enhancement proposed by Pramunendar et al. [11] by proposing a gray-level co-occurrence matrix (GLCM) feature extraction fed into the back-propagation neural network (BPNN) with contrast-adaptive color correction technique (NCACC) for Underwater Fish Classification. In the case of medical images, Reddy et al. [12] proposed Integrated Guided Nonlinear Anisotropic Filtering pada CT and MR images. Another preprocessing technique that can be used to reduce noise is autoencoder. Gondara [13] once used Convolutional Autoencoder to denoise medical images. Ke et al. [14] proposed an end-to-end automatic image annotation model based on deep Convolutional Neural Network (E2E-DCNN) by generating new images based on original image. The same technique has been applied by Li and Yeh [15] to deal with the problem of data limitations and variations. Various studies have applied preprocessing techniques to improve image quality. However, they have not used preprocessing techniques to enhance image recognition performance.

Several researchers have applied deep learning to the various characters recognition. Rabby et al. [16] examined the recognition of Banglas handwritten characters using CNN. This study only recognizes the basic characters, numbers, modifiers, and Bangla combined characters. Sonawane and Shelke [17] applied transfer learning using Alexnet to recognize Devanagari handwritten characters. Pragathi et al. [18] used Concolutional Neural Network to recognize handwritten Tamil characters. Related research has been conducted by Elleuch et al. [19] that proposed a new SVM-based design of the CNN architecture with Dropout for the Arabic characters recognition. They explained a new model of exploration that focuses on integrating two classifiers, namely CNN and SVM, for the recognition of Offline Arabic Handwriting (OAHR). In the case of Chinese characters recognition, Gan et al. [20] compared Recurrent Neural Network (RNN) and Convolutional Neural Network 1-D for Online Handwritten Chinese Character (OLHCCR) dataset recognition. Related research has been done by Zhang et al. [21] by proposing a combination of Resnet and Center Loss Based Metric Learning to improve the accuracy of recognition.

Transfer learning is a technique that is widely applied

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Figure 1. Balinese *Lontar* Image

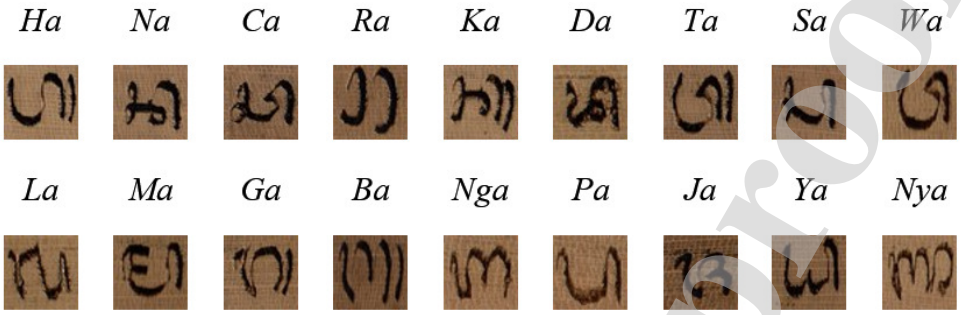


Figure 2. Balinese characters

to improve recognition performance. Various studies applied transfer learning on the recognition of various objects. Shallu and Mehra [22] proposed a transfer learning method for breast cancer classification using VGG16, VGG19, and ResNet50 by examining the effect of training-testing data size on network performance. Falconi et al. [23] applied MobileNet, Nasnet, InceptionV3, and Resnet50 to classify abnormalities in breast mammogram images to determine each models performance. Swasono et al. [24] applied the pre-trained model to classify tobacco leaf pests. The study used the VGG16 model on 1500 tobacco images. Related studies proposed by Vaka et al. [25] proposed Deep Neural Network with Support Value (DNNS) to produce better quality images and to fix other performance parameters and evaluate the efficiency and performance. Hanif and Bilal [26] proposed a competitive residual network (CoRN) for image classification. The proposed network is composed of residual units having two identical blocks each containing convolutional filters, batch normalization, and a maxout unit to enables the competition among the convolutional filters and reduces the dimensionality of the convolutional layer.

3. Methodology

We proposed a new Multi Augmentation Technique for data augmentation using a combination of Adaptive Gaussian Thresholding and Convolutional Autoencoder (MAT-AGCA) to overcome the limited amount of data in the dataset. This method can provide data variations by reducing noise in the original image to improve recognition performance. Fig. 3 shows our proposed method. The first phase is MAT-AGCA which has two sub-processes, i.e. AGT and CAE. This phase used original dataset as input to produce a Handwritten Balinese Characters of *Lontar* (HBCL) dataset by combining MAT-AGCA and original dataset. The second phase is the training process to produce a model using CNN. In this phase, we applied three pre-trained models to practice the recognition model. The third phase is the Balinese characters recognition process to test recognition performance. The following is an explanation of material and methods in this study.

3.1. Balinese characters on lontar manuscript

Balinese characters are an ancient characters used to write cultural manuscripts in Bali. Balinese characters consists of 18 classes called *Wianjana* characters. The *lontar* manuscript is made from dried palm leaves. Balinese characters on a *Lontar* manuscript are written using a knife called *pengrupak*. Knife strokes form Balinese characters which are then colored using candlenut powder. Fig. 1 shows the Balinese cultural manuscript made from palm leaves. Balinese characters in the Balinese cultural manuscript has unique characteristics because it is the result of hand scratches using a *pengrupak*. It causes inconsistent characters shapes that are more difficult to recognize than handwritten characters on paper or other media, for example the *Ra* character in Fig. 2. The

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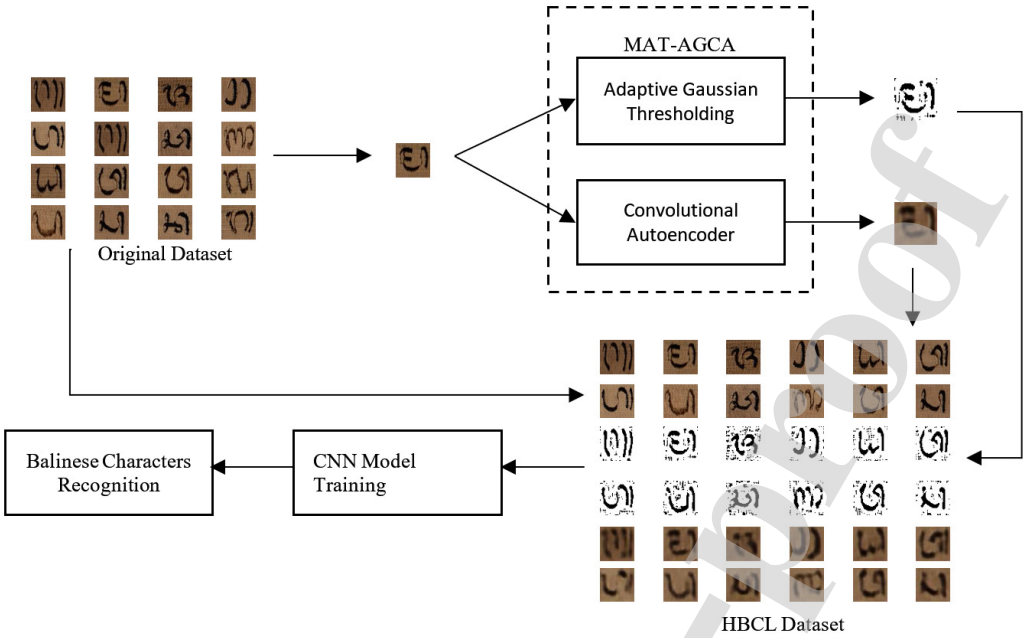


Figure 3. Proposed method: multi augmentation techniques-adaptive gaussian convolutional autoencoder (MAT-AGCA)

use of candlenut powder as a coloring gives noise to the *Lontar* manuscripts, thus it becomes a challenge in the recognition process. For example, the *Da* in Fig. 2 shows the noise caused by candlenut powder. Besides, cultural manuscripts made from palm leaves cause leaf fibers to become noise in the acquisition process of Balinese characters.

The limited variation of Balinese characters is challenging in recognizing Balinese character. Balinese characters have inconsistent and varied forms of writing. Therefore, we collected Balinese characters from 50 lontar manuscript writers in Bali to enrich the variations. We used this dataset as a basis for the data augmentation process to improve recognition performance. Fig. 1 shows the *lontar* manuscript used in this study. We acquired the image of Balinese characters on *lontar* manuscript using a scanner. Then, we cut each Balinese characters image with a size of 48×48 pixel. This process produces 1195 images of Balinese characters as the original dataset. For example, the results of image acquisition can be seen in Fig. 2.

3.2. *Multi augmentation techniques-adaptive gaussian convolutional autoencoder (MAT-AGCA)*

We proposed the Multi Augmentation Technique (MAT-AGCA) for data augmentation using a combination of Adaptive Gaussian Thresholding (AGT) and Convolutional Autoencoder (CAE). MAT-AGCA applied AGT and CAE to increase data variation in the dataset by improving image quality. Adaptive Gaussian Thresholding is an adaptive thresholding technique using a Gaussian filter depending on the value of standard deviation. In one dimension, the gaussian function is:

$$G(x, y) = \frac{1}{\sqrt{2\pi\sigma^2}} e \frac{x^2}{2\sigma^2} \quad (1)$$

This filter serves to remove high-frequency components in the image. The σ is the standard deviation and pixels at the center (x,y), where x is a gaussian function that determines the threshold result, and y is a constant with the square root of negative one. This process uses the input image shown in Fig. 2, which is the original image from the image acquisition results. Fig. 5(a) shows the results of a gaussian filter. Window size is a parameter that affects the quality of the output image on AGT. Experiments were carried out with variations in window size: 3×3, 5×5, 7×7, 11×11, and 13×13. Handwritten Balinese characters have various writing styles according to the writer. This is a challenge to recognize the handwritten Balinese characters on *lontar*. *Lontar* manuscripts are written using a Pengrupak, thus resulting in the Balinese characters writing becomes varied. Therefore, to be able to recognize variations of Balinese characters found in *lontar*, specific techniques are needed to pro-

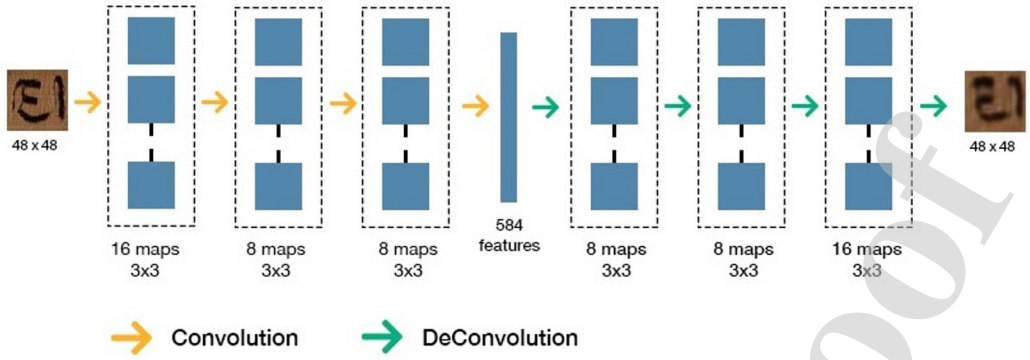


Figure 4. Convolutional autoencoder architecture

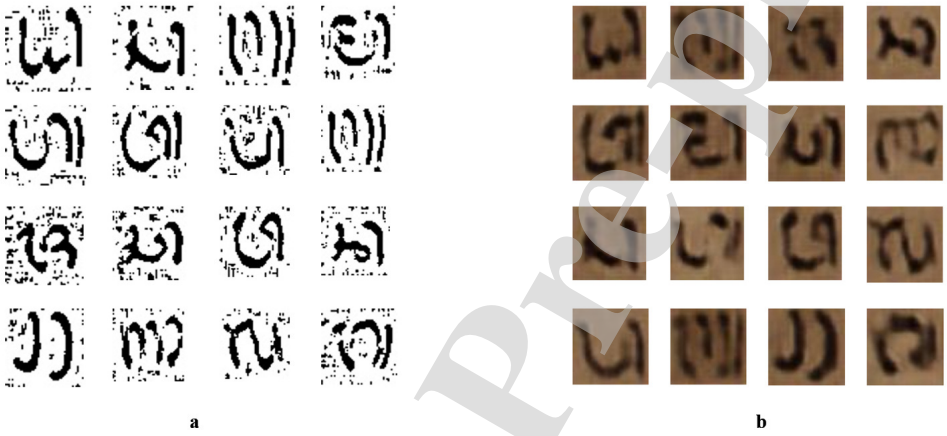


Figure 5. Augmentation aesult of MAT-AGCA: (a) AGT and (b) CAE

vide variations on the dataset. In addition, the limitations of the original dataset and the similarity of scripts between classes became an exciting challenge; hence it is necessary to do the data variation techniques. We used Convolution Autoencoder (CAE) to produce Balinese characters variations in the original data. This technique is used to overcome the limited number of data variations in Balinese character images. CAE generally has the same steps as the classic autoencoder, but the neural network encoder and decoder models use CNN. For example, in the encoder section, the feature vector generation process is the same as an ordinary CNN model. The end of the encoder convolution layer is then flattened to produce a feature vector. CAE has differences in the decoder section. After the feature vector is rearranged into 2D, it takes an upsampling or transposes the convolution technique to do the deconvolutional process. Compared to classic autoencoder, CAE is better used for image processing because it utilizes the full capability of convolutional neural networks to exploit image structures [13]. Fig. 4 shows the CAE architecture used in this study. The input image has a size of 48×48, and the output has the same size. Yellow lines indicate convolutional processes that produce features. In the encoder stage, the input data passes through 3 convolutional layers (Conv2D) and 3 pooling layers (MaxPooling2D). On the convolution layer, we use a 3×3 kernel to help in organizing the network by forcing it to have decomposition on a large receptive field. Green lines are deconvolutional from features that produce output images of the same size. The deconvolutional process uses upsampling techniques. We applied CAE with a different number of epochs variation e.g. 50, 100, and 150 epoch to produce the best augmentation result.

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Table 1
Experimental scenarios

Scenario	Number of Images in Dataset	Data Distribution		
		Training	Validation	Testing
Without Augmentation	1197	945	144	108
Geometric Transformation	3159	2835	216	108
Convolutional Autoencoder	3159	2835	216	108
Adaptive Gaussian Thresholding	3159	2835	216	108
Proposed Method	3159	2835	216	108

3.3. Experimental Scenarios

CNN is an artificial neural network that is effectively used in image processing. CNN is an artificial neural network architecture through additional layer structures that are used as a substitute for hidden layers. The convolutional layer uses a small weighing kernel to produce visual features [20] . CNN takes a long time, depending on the amount of data used to form the model. The more data is processed, the longer the time is needed. To shorten the training time, we used transfer learning techniques using pre-trained models. In addition to shortening the training time, pre-trained models can improve accuracy and model performance.

We compiled Balinese characters data from ancient lontar manuscripts and 50 lontar writers. 50 lontar writers aim to provide variations in the writing style of Balinese characters. Based on this compilation process, we built a dataset of Balinese characters totaling 1197 characters consisting of 18 classes. In the experimental scenario, we perform five scenarios. The first scenario uses a dataset without applying augmentation. The second scenario applies Geometric Transformation (GT) by using rotation, height shift, and width shift transformations. The third scenario applies a convolutional autoencoder to add variation to the dataset. The fourth scenario applies Adaptive Gaussian Thresholding to the dataset to improve image quality by reducing noise. The fifth scenario applies our proposed method for augmentation. We use five CNN pre-trained models, i.e. InceptionResnetV2, DenseNet169, ResNet152V2, VGG19, and MobileNetV2 to test the recognition performance in each scenario. Table 1 shows the experimental scenarios of this study.

4. Results and discussions

4.1. Data augmentation

In the first scenario, we did not apply data augmentation to the dataset. We use this dataset as a basis for determining the influence of applying data augmentation in each scenario. The dataset without augmentation consists of 1,197 images of Balinese characters composed of ancient lontar manuscripts and 50 lontar writers. This dataset consists of 18 Balinese character classes.

The second scenario applies Geometric Transformation as a data augmentation technique. We use three transformations, i.e. rotation, width shift, and height shift. Geometric Transformation is applied to the dataset used in the first scenario. Based on the augmentation process, the total data in the second scenario becomes 3,159 images of Balinese characters. We implemented the CAE in the third scenario to add variation to the dataset. CAE has a convolution and deconvolution process. Each of which consists of 3 convolution layers with a kernel size of 3×3. We implemented CAE with 3 variations of epoch, i.e. 50, 100, and 150. In this experiment, the best results for CAE were obtained in 100 epochs. AGT applied in the fourth scenario. This preprocessing technique is used to improve image quality by reducing noise in the image. AGT augmentation results are influenced by the window size. We use various window sizes: 3×3, 5×5, 7×7, 11×11, and 13×13 to get the best augmentation results. In the last scenario, we implemented our proposed method. MAT-AGCA produced 3,159 images from a combination of style variations from CAE and image quality enhancement from AGT. We propose the MAT-AGCA augmented dataset as a new Balinese character dataset called Handwritten Balinese Characters of Lontar Manuscript (HBCL).

4.2. Model training using InceptionResNetV2

We conducted the first test using InceptionResNetV2. In the dataset without augmentation, InceptionResNetV2 yielded 54.7% training accuracy, 21.2% validation and testing accuracy. It shows that the model is overfitting due to the small amount of data. Further, we applied Geometric Transformation as data augmentation with three operations, i.e. rotation, width, and height shifting that can improve performance by 10.2%. In this scenario, we also applied the convolutional autoencoder, adaptive

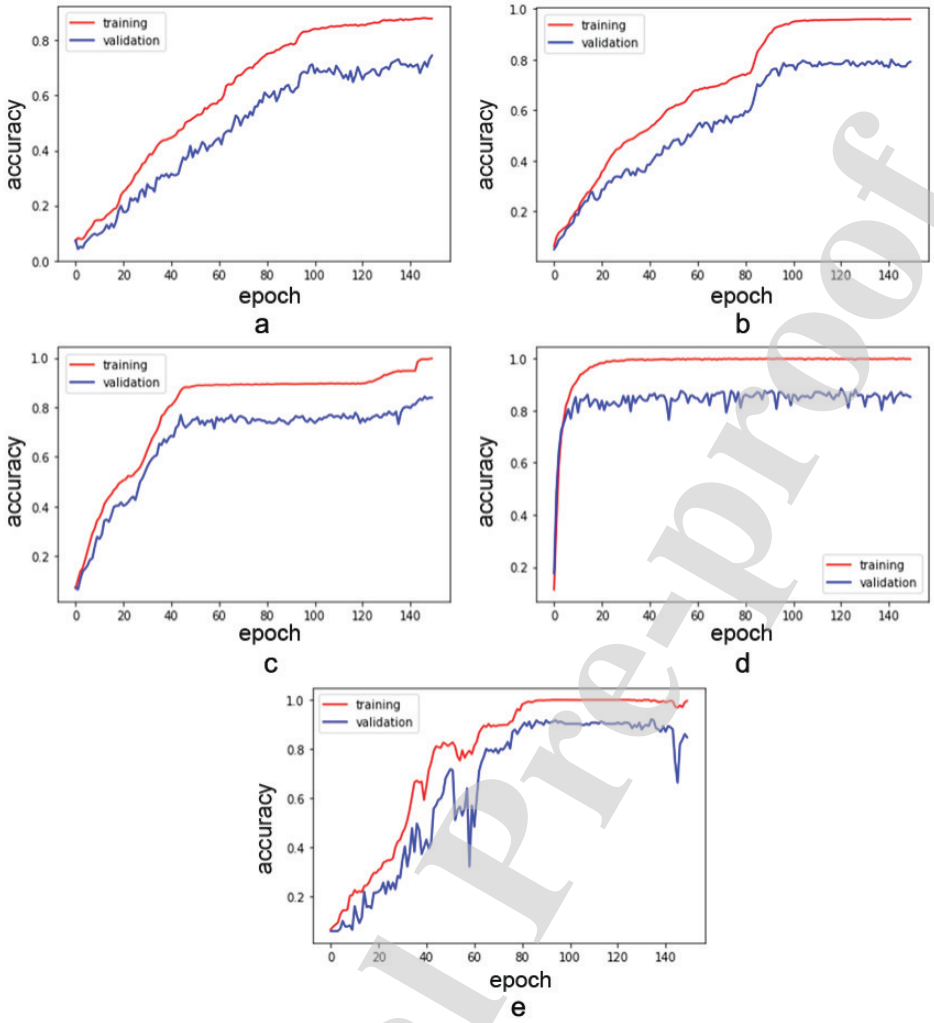


Figure 6. MAT-AGCA performance: (a) InceptionResNetV2, (b) DenseNet169 (c) ResNet152V2, (d) VGG19, (e) MobileNetV2

gaussian thresholding, and our proposed method. The experimental results in this scenario, CAE and AGT increase performance by 28.8% and 39.9% respectively. Our proposed method improve performance by 61.2% with 82.4% testing accuracy.

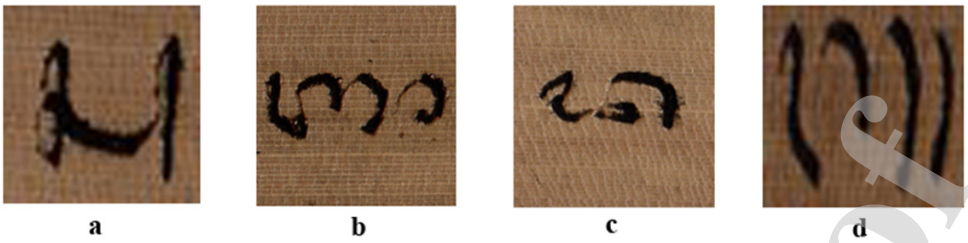
4.3. Model training using DenseNet169

In the second test, we used DenseNet169 as the pre-trained model. We tested five types of datasets based on experimental scenarios to determine the performance of our proposed method. In the dataset without augmentation, the model shows overfitting because there is a 40% gap between training and validation. After applying data augmentation using Geometric Transformation, CAE, AGT, and our proposed method, our proposed method achieved the best model by 50.9% performance improvements with 90.7% testing accuracy.

4.4. Model training using ResNet152V2

In the third test, we used ResNet152V2 as the pre-trained model. In the dataset without augmentation, the model yielded 44.4% testing accuracy for 108 Balinese character images. It shows that the model is only able to recognize less than half of the tested data . After applying the four types of data augmentation includes our proposed method, the model produce better performance by 95.3% testing accuracy. These results show that our proposed method improve the performance of

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a b c d

Figure 7. Failed recognized characters on the previous method: (a) *sa* character, (b) *nya* character, (c) *da* character, (d) *ba* character

Table 2
Performance comparison on five CNN pre-trained models

Pre-trained Models		Without Augmentation	Augmentation Methods			Proposed Method
			Geometric Transformation [27] [28] [29] [30]	Convolutional Autoencoder [31] [32] [33]	Adaptive Gaussian Thresholding [8] [34]	
InceptionResNetV2	Train	54.70%	60.10%	73.26%	92.26%	91.88%
	Val	21.29%	33.33%	50.00%	61.11%	82.40%
	Acc	21.29%	31.48%	50.00%	61.11%	**82.40%**
DenseNet169	Train	98.80%	93.43%	84.23%	94.42%	95.94%
	Val	39.81%	61.11%	63.88%	88.88%	90.74%
	Acc	39.81%	57.40%	63.88%	88.88%	**90.74%**
ResNet152V2	Train	97.84%	93.82%	98.73%	100%	99.68%
	Val	44.44%	50.92%	73.61%	97.22%	95.37%
	Acc	44.44%	50.92%	78.70%	90.74%	**95.37%**
VGG19	Train	100%	99.15%	100%	99.85%	99.82%
	Val	61.11%	67.59%	89.81%	93.51%	90.74%
	Acc	61.11%	70.37%	89.81%	93.51%	**96.29%**
MobileNetV2	Train	53.33%	75.76%	99.36%	98.87%	99.61%
	Val	41.66%	48.14%	88.88%	93.51%	96.29%
	Acc	40.74%	54.62%	90.74%	92.59%	**96.29%**

the ResNet152V2 model by 50.9%. Compared to Geometric Transformation, CAE, and AGT, which only improve the model's performance under 50%.

4.5. Model training using VGG19

We used VGG19 as the pre-trained model for the fourth test. The models produced on the four types of augmentation techniques show varying performance improvements. Geometric Transformation produce 9.2% performance improvements. CAE showed a better model with 28.7% improvement, AGT improve model's performace by of 32.4%, and our proposed method resulted in the highest performance improvements by 35.1%.

4.6. Model training using MobileNetV2

We also applied MobileNetV2 that has the least number of parameters. This scenario aims to determine our proposed method performance on the pre-trained model with few parameters. The results show that our proposed method can improve the model performance by 55.5% over Geometric Transformation, CAE, and AGT. Our proposed method is still able to provide improved model performance on the small dataset using a pre-trained model, which has a small number of parameters.

4.7. Model Performance comparison

In the conducted experiments, the application of MAT-AGCA in data augmentation is influenced by the epoch and window size variation parameters that affect the data augmentation results quality. We applied the different number of epochs variation e.g 50, 100, and 150 epochs with window size variation e.g. 3×3, 5×5, 7×7, 11×11, and 13×13 to produce the best augmentation result. In the case of Balinese characters, the best augmentation

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image quality is produced using 100 epochs and a window size of 7×7. Fig. 6 shows the performance of MAT-AGCA using five pre-trained models.

Based on the experimental results using four types of augmentation techniques with five pre-trained models, our proposed method can produce better model performance. Experiments are carried out using the RMSProp optimizer with a learning rate of 0.00001 and softmax activation for 150 epochs. In the experiment using the previous methods, there were still several Balinese characters that failed to be recognized, as shown in Fig. 7 . It due to the Balinese characters characteristics in the lontar manuscript, which have been described in Section 3. For example, there is a broken pattern in the *Nya* character causing the wrong recognition shown in Fig. 7(b). Our proposed method, MAT-AGCA can solve this problem by increasing the variety of data while improving image quality.

In the second scenario, we applied the data augmentation used in [27] [28] [29] [30] that applies Geometric Transformation as data augmentation. The results show that our proposed method produces better performance in the case of Balinese characters with an improvements in model performance by up to 55.5%. In the case of Balinese characters in the lontar manuscript that has different characteristics from other characters, the application of augmentation Geometric Transformation cannot produce a new character image that represents the characteristics of Balinese characters writing. In the third scenario, adding data variations implements the convolutional autoencoder used in [31] [32] [33]. CAE only provides variations by reconstructing the image pattern using a convolutional network. CAE generates a new image based on a variety of writing styles without increasing image quality, therefore the generated images unable to represent the characteristics of Balinese character images in lontar manuscripts which have noise. Compared to CAE, our proposed method produces a better average model performance on the five pre-trained models by 17.5%. The application of AGT as data augmentation is used in [8] [34] by providing data variations based on noise reduction produces an average performance of the five pre-trained models of 85.3%. AGT can provide better model performance than Geometric Transformation and CAE because it produces augmented images with noise reduction. Compared to AGT, our proposed method can produce better performance with the highest testing accuracy up to 96.2%. Table 2 show performance comparison of our proposed method and other studies using five CNN architectures

Based on experimental results, our proposed method gives better performance and suitable for Balinese character recognition. The experimental results prove that data augmentation using MAT-AGCA based on noise reduction and style variations significantly improves recognition performance. Our proposed method could produce a better performance on VGG19 that has a large number of parameters and MobileNetV2 that has a few parameters.

5. Conclusions and future work

In this paper, we proposed MAT-AGCA as data augmentation to overcome the problem of recognition on datasets with limited data. We presented a solution to overcome the noise problem in Balinese character images in *Lontar* manuscripts. The proposed method can reduce noise and provides image variations to improve recognition performance. Conventional augmentation only provides data variations by applying Geometric Transformation, such as rotation and scaling, without reducing noise. Our proposed method can reduce noise; thus, it can increase recognition accuracy by enriching image variations on the dataset. Based on the five experiments using five pre-trained models, MAT-AGCA achieved the best performance with 96.29% accuracy, outperforms existing augmentation method. By applying MAT-AGCA, average recognition accuracy on five pre-trained models increased by 50.7%. In this study, we produced the Handwritten Balinese Characters of *Lontar* (HBCL) dataset using MAT-AGCA. The HBCL dataset is compiled from augmentation results in Balinese character images through image quality enhancement and variations representing various handwritten styles on *Lontar* manuscripts. This dataset consists of 18 class of *Wianjana* script as the main character without *Gantungan* and *Gempelan* scripts.

As future work, we would like to develop a new architecture to recognize the *Gantungan* and *Gempelan* scripts with the specific rules of reading sentences in Balinese script, thus it can recognize a whole Balinese script. Specific methods are still required to overcome the challenges that still exist in reading Balinese characters.

Conflict of interest

All the authors declare that there is no conflict of interest regarding the publication of this paper.

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AUTHORSHIP STATEMENT

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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